

APPROXIMABILITY FOR QUASI-COMPACT QUASI-SEPARATED SCHEMES

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ABSTRACT. Let X be a quasi-compact quasi-separated scheme. We give necessary and sufficient conditions for the approximability, in the sense of Neeman, of the derived ∞ -category of quasi-coherent sheaves $\mathrm{QCoh}(X)$. We consequently provide a large class of non-separated schemes whose category $\mathrm{QCoh}(X)$ is approximable. Furthermore, if $Z \subseteq X$ is a closed subscheme with quasi-compact open complement, we show that approximability of the stable ∞ -category $\mathrm{QCoh}_Z(X)$ forces the underlying topological space $|Z|$ to be open. More precisely, $\mathrm{QCoh}_Z(X)$ is approximable if and only if $|Z|$ is open and the corresponding open-and-closed subscheme $W \subseteq X$ has approximable category $\mathrm{QCoh}(W)$.

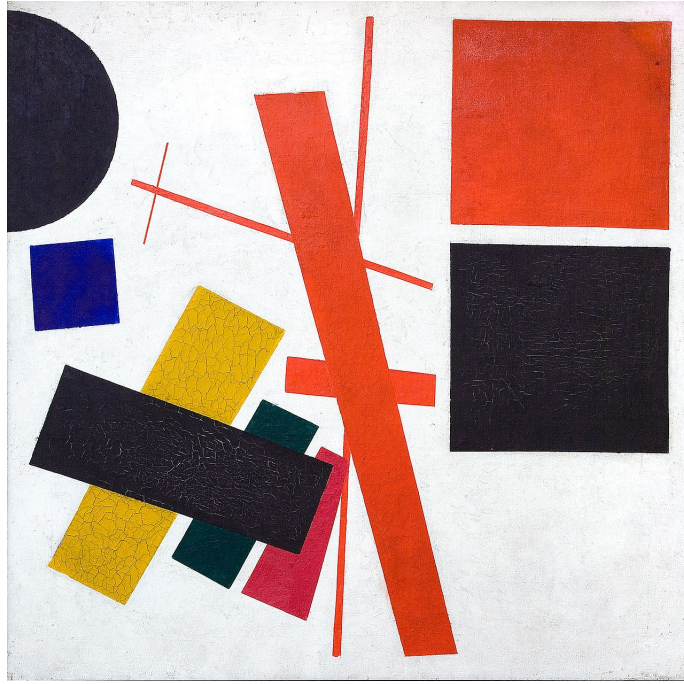


FIGURE 1. *Suprematism. Abstract Composition*, Kazimir Malevich, 1915.

1. INTRODUCTION

Compactly generated stable categories are ubiquitous in geometry and homotopy theory, and many finiteness questions about them can be understood as approximation problems. In such categories every object can be recovered from compact objects by filtered colimits. While this is a powerful finiteness statement, it does not say how efficient this recovery is. A t -structure makes this question sharper, since it allows one to measure the error of an approximation. In other words, compact objects provide the basic building blocks, while the t -structure controls the quality of the approximation.

Neeman's notion of *weak approximability* [Nee25, Definition 0.25] makes this idea precise since it does not merely ask that a compact generator generate the category; it asks that connective objects admit approximations by arbitrary coproducts of bounded shifts of one compact generator. For applications, weak approximability is not enough; one needs the strength of *approximability*. The difference between the former and the latter is substantial, in that in the latter connective objects can be approximated by objects built from bounded shifts of a compact generator using arbitrary coproducts but finitely many extensions.

Regarding examples, Neeman showed in [Nee25, Example 4.6] that the (homotopy category of the) derived stable ∞ -category of quasi-coherent sheaves $\mathrm{QCoh}(X)$ is weakly approximable for every quasi-compact quasi-separated scheme X and approximable if X is furthermore separated. Our first result extends (and characterises) approximability of schemes with a particularly nice basis of affine opens.

Theorem. Let X be a quasi-compact quasi-separated scheme. Then $\mathrm{QCoh}(X)$ is approximable if and only if it admits a basis of affine extremely compactly bounded opens.

We refer the reader to Definition 3.1 for the definition of extremely compactly bounded morphisms. Roughly, it asks the pushforward applied to a compact object to be generated from a compact object using coproducts, retracts and a finite number of extensions and shifts. The necessity of the above is then proved in Proposition 3.6. The hardest part is to prove sufficiency (Theorem 4.3), and the main idea is to show that approximability is Zariski local, so that the statement easily follows from the approximability of derived categories of modules. We point out that the methods used to prove the first theorem should be general enough to apply to more general categories than quasi-coherent sheaves on a scheme and never ask the target to be weakly approximable.

The examples in section 5 show that the criterion is flexible enough to produce genuinely non-separated schemes with approximable derived categories. In particular, it applies to schemes obtained by gluing two separated schemes along a sufficiently well-behaved quasi-compact open subscheme, and to certain schematic quotients arising from hyperbolic actions of G_m . This includes, for instance, the affine line with two origins.

Another example of weakly approximable category is given by categories of quasi-coherent sheaves with support $\mathrm{QCoh}_Z(X)$ for X quasi-compact quasi-separated and $Z \subseteq X$ closed subscheme with quasi-compact open complement. Our second result shows that for these categories approximability is a very rigid condition: it occurs only in the evident split case.

Theorem. Let X be a quasi-compact quasi-separated scheme and let Z be a closed subscheme of X with quasi-compact open complement. Then $\mathrm{QCoh}_Z(X)$ is approximable if and only if the underlying topological space $|Z|$ is open and the open subscheme $W \subseteq X$ with underlying topological space $|Z|$ has approximable category $\mathrm{QCoh}(W)$. In this case, $\mathrm{QCoh}_Z(X) \simeq \mathrm{QCoh}(W)$.

The proof uses the standard recollement $\mathrm{QCoh}_Z(X) \hookrightarrow \mathrm{QCoh}(X) \rightarrow \mathrm{QCoh}(U)$, where $U = X \setminus Z$. If $|Z|$ is open, then X splits as the disjoint union of U and the corresponding open subscheme W , and the claim follows since $\mathrm{QCoh}_Z(X)$ identifies with $\mathrm{QCoh}(W)$. Conversely, assume that $\mathrm{QCoh}_Z(X)$ is approximable. A compact generator of $\mathrm{QCoh}_Z(X)$ is perfect on X , hence killed by a fixed power of an ideal defining $|Z|$. Uniform approximability then forces the same power of the ideal to be idempotent,

making $|Z|$ open and closed. We would like to point out that this idea for the proof was contained in [Nee24, Remark 8.1].

Organization. The note is organized as follows. In section 2 we recall the definition of approximability and the basic examples. In section 3 we introduce the notion of extremely compactly bounded morphism and prove some properties. We then use this definition in section 4 to prove approximability of derived ∞ -categories of quasi-coherent sheaves on a scheme by Zariski descent. In section 5 we discuss some examples. In section 6 we analyse categories of sheaves with support. We recall the recollement of derived categories associated to an open-closed decomposition of a scheme and use it to produce a characterization of those closed subschemes which are also open subschemes. Finally, in section 7 we prove the characterization of approximability for sheaves with support in a closed.

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Convention. We use the following conventions.

Notation 1.1. We freely use the language of stable ∞ -categories (also known as stable *categories*) and t -structures on them, which we grade homologically. If $\mathcal{C}$ is a stable ∞ -category, we let $\mathrm{hom}_{\mathcal{C}}(-, -)$ denote the mapping spectrum. We denote by Sp the stable ∞ -category of spectra and we equip it with the standard t -structure. We stress that the connective half is closed under arbitrary products.

Notation 1.2. If A is a ring, we will denote by Mod_A the derived stable ∞ -category of modules over A . Given a quasi-compact quasi-separated scheme X , we denote by $\mathrm{QCoh}(X)$ the derived stable ∞ -category of quasi-coherent sheaves on X and by $\mathrm{Perf}(X)$ the corresponding category of compact objects. We equip $\mathrm{QCoh}(X)$ with the standard t -structure. We denote by $|X|$ the underlying topological space of X . We short the notation $\mathrm{hom}_{\mathrm{QCoh}(X)}(-, -)$ to $\mathrm{hom}_X(-, -)$. All maps of schemes are quasi-compact quasi-separated.

Notation 1.3. Let X be a quasi-compact quasi-separated scheme and let $j : U \hookrightarrow X$ and $i : Z \hookrightarrow X$ be a decomposition of X into a quasi-compact open U and closed Z .

- (1) Consider the adjoint functors $j^* : \mathrm{QCoh}(X) \rightleftarrows \mathrm{QCoh}(U) : j_*$. Then j^* is t -exact and j_* is fully-faithful and right t -exact up to a shift.
- (2) Consider the adjoint functors $i^* : \mathrm{QCoh}(X) \rightleftarrows \mathrm{QCoh}(Z) : i_*$. Then i_* is t -exact but not fully-faithful in general. Furthermore, the pullback functor i^* is not left t -exact, since the inclusion $i : Z \hookrightarrow X$ is not of finite tor-amplitude.

We let $\mathrm{QCoh}_Z(X)$ denote the full subcategory of $\mathrm{QCoh}(X)$ spanned by those quasi-coherent sheaves $F \in \mathrm{QCoh}(X)$ such that $\pi_n(F) \in \mathrm{QCoh}(X)^\heartsuit$ is supported on Z (that is, their restriction to U vanishes).

2. THE DEFINITION OF APPROXIMABILITY

The definition of approximability requires some more notation (see [Nee25, Reminder 0.12]).

Notation 2.1. Let $\mathcal{C} \in \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ be a presentable stable category. Let $S \subseteq \mathcal{C}$ be a collection of objects. Given two (possibly infinite) integers $-\infty \leq a \leq b \leq +\infty$, we let $S[a, b]$ denote the full subcategory of $\mathcal{C}$ spanned by the $\Sigma^{-i}s$ for $s \in S$ and $a \leq i \leq b$. Given $n \in \mathbb{N}$ we define inductively the full subcategories $\mathrm{Thick}_n(S)$ as follows. We let $\mathrm{Thick}_1(S)$ be the smallest subcategory containing arbitrary coproducts of S and closed under retracts and $\mathrm{Thick}_n(S)$ to be the smallest subcategory closed under retract on those $y \in \mathcal{C}$ such that there exists an exact sequence $x \rightarrow y \rightarrow z$ with $x \in \mathrm{Thick}_1(S)$ and $z \in \mathrm{Thick}_{n-1}(S)$. We let $\mathrm{Thick}(S)$ be smallest full subcategory closed under coproducts, extensions and retracts containing S . Notice that this is not the union of all the $\mathrm{Thick}_n(S)$. In the case where $S = \{G\}$ is a single object, we omit the parentheses. We also omit the brackets $[a, b]$ if a and b are infinite. We will refer to $\mathrm{Thick}_n(G[-n, n])$ as the n -th *proxy small closure* of G . We will also refer to objects therein as *proxy small*.

Neeman gave in [Nee25, Definition 0.25] the definition of a (weakly) approximable triangulated category. In ∞ -categorical terms, it translates to the following.

Definition 2.2. Let $\mathcal{C} \in \text{Pr}_{\text{st}}^{L, \omega}$ be a compactly generated stable category. We will say that $\mathcal{C}$ is *weakly approximable* if there exist a compact generator $G \in \mathcal{C}^\omega$, a t -structure $(\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0})$ and an integer $A \in \mathbb{N}$ such that:

- (1) It is $\Sigma^A G \in \mathcal{C}_{\geq 0}$ and $\text{hom}_{\mathcal{C}}(G, \mathcal{C}_{\geq A}) \in \text{Sp}_{\geq 1}$.
- (2) For every connective object $x \in \mathcal{C}_{\geq 0}$ there exists an exact sequence $p \rightarrow x \rightarrow e$ in which $p \in \text{Thick}(G[-A, A])$ and $e \in \mathcal{C}_{\geq 1}$.

We will furthermore say that $\mathcal{C}$ is *approximable* if condition (2) can be strengthened to:

- (3) For every connective object $x \in \mathcal{C}_{\geq 0}$ there exists an exact sequence $p \rightarrow x \rightarrow e$ in which $p \in \text{Thick}_A(G[-A, A])$ and $e \in \mathcal{C}_{\geq 1}$.

Remark 2.3. Let $\mathcal{C} \in \text{Pr}_{\text{st}}^{L, \omega}$ be a compactly generated stable category with a compact generator and a t -structure. Notice that if $A \in \mathbb{N}$ shows (weakly) approximability, then every index $A' \geq A$ will show (weakly) approximability.

Remark 2.4. The reason behind the nomenclature “approximability” is well explained in [Nee22, Remark 2.5 and Fact 2.6], in that condition (3) can be used to produce arbitrarily good approximations. To be precise, let $\mathcal{C} \in \text{Pr}_{\text{st}}^{L, \omega}$ be an approximable category and pick a compact generator G together with a t -structure and an integer $A \in \mathbb{N}$ satisfying the axioms. Let $x \in \mathcal{C}_{\geq 0}$ be connective. Then for every $n \geq 1$ there exists an exact sequence $p_n \rightarrow x \rightarrow e_n$ in which $p_n \in \text{Thick}_{nA}(G[-A - n, A + n])$ and $e_n \in \mathcal{C}_{\geq n}$. This can be shown by induction. If $\mathcal{C}$ is assumed only to be weakly approximable, the same statement holds, but $p_n \in \text{Thick}(G[-A - n, A + n])$ has no bound on the number of extensions.

The main examples of (weakly) approximable categories are given by derived categories of schemes.

Example 2.5. Let X be a quasi-compact quasi-separated scheme. Then [Nee25, Example 4.6.] shows that $\text{QCoh}(X)$ is weakly approximable. More precisely, the compact generator is given by the construction of Bondal and Van den Bergh [BdB02, Theorem 3.1.1(ii)] and the t -structure may be chosen to be the standard one. The existence of the required index is shown in [Nee25, Lemma 4.5]. If X is moreover separated, then $\text{QCoh}(X)$ is approximable.

Example 2.6. Let X be a quasi-compact quasi-separated scheme and let $Z \subseteq X$ be a closed subscheme with quasi-compact open complement. Then [Nee24, Theorem 3.2] states that $\text{QCoh}_Z(X)$ is weakly approximable and the t -structure used to prove weakly approximability is equivalent to the standard t -structure.

3. COMPACTLY BOUNDED MORPHISMS

Inspired by [DLR25, Definition 4.1], we give the following.

Definition 3.1. Let $f : X \rightarrow Y$ be a morphism of quasi-compact quasi-separated schemes. We will say that f is *extremely compactly bounded* if there exists an integer $n \in \mathbb{N}$ such that, for every $P \in \text{Perf}(X)$, there exists $Q \in \text{Perf}(Y)$ such that $f_*(P) \in \text{Thick}_n(Q[-n, n])$.

Example 3.2. Let X be a quasi-compact quasi-separated scheme and $j : U \hookrightarrow X$ be the inclusion of a quasi-compact open. To show that j is extremely compactly bounded, via the projection formula and Thomason-Trobaugh, it suffices to show that there exists an integer $n \in \mathbb{N}$ such that there exists a compact object $Q \in \text{Perf}(X)$ for which $j_*(\mathcal{O}_U) \in \text{Thick}_n(Q[-n, n])$. This was observed in [DLR25, Example 4.2]. In particular, if X is furthermore separated, then [Nee24, Theorem 6.2] shows that j is extremely compactly bounded.

Lemma 3.3. Composition of extremely compactly bounded is extremely compactly bounded. If the composition of two morphisms is extremely compactly bounded and the second map is a quasi-compact open inclusion, then the first map is extremely compactly bounded.

Proof. Let X be a quasi-compact quasi-separated scheme $i : V \rightarrow U$ and $j : U \rightarrow X$ be two quasi-compact opens. The first claim is obvious, so assume that the composition $j \circ i$ is compactly bounded. Let $P \in \text{Perf}(V)$. Then there exists $Q \in \text{Perf}(X)$ and an integer $n \in \mathbb{N}$ such that $j_* i_*(P) \in \text{Thick}_n(Q[-n, n])$. Then $i_*(P) \simeq j^* j_* i_*(P) \in j^*(\text{Thick}_n(Q[-n, n])) = \text{Thick}_n(j^* Q([-n, n]))$ and the claim follows since j^* preserves compact objects (and the fully-faithfulness of j_*). $\square$

The following result is extremely useful to produce extremely compactly bounded open immersions:

Lemma 3.4. Let X be a quasi-compact quasi-separated scheme. If X admits a basis of extremely compactly bounded quasi-compact opens, then every quasi-compact open immersion is extremely compactly bounded.

Proof. Let $\mathcal{P}$ the class of extremely compactly bounded quasi-compact opens of X . Then $\mathcal{P}$ is closed under finite intersections. Indeed let $j_U : U \hookrightarrow X$ and $j_V : V \hookrightarrow X$ be in $\mathcal{P}$ and set $j_W W = U \cap V \hookrightarrow X$ for their intersection. Then there exist integers $n_U, n_V \in \mathbb{N}$ and perfect complexes $Q_U, Q_V \in \text{Perf}(X)$ such that

$$(j_U)_* \mathcal{O}_U \in \text{Thick}_{n_U}(Q_U[-n_U, n_U]), \quad (j_V)_* \mathcal{O}_V \in \text{Thick}_{n_V}(Q_V[-n_V, n_V]).$$

Using the projection formula and flat base change shows that $(j_W)_* \mathcal{O}_W \simeq (j_U)_*(\mathcal{O}_U) \otimes_{\mathcal{O}_X} (j_V)_*(\mathcal{O}_V)$ and since tensoring is exact and preserves coproducts and retracts, it follows that $(j_W)_* \mathcal{O}_W \in \text{Thick}_N((Q_U \otimes Q_V)[-N, N])$ for some integer $N \in \mathbb{N}$, depending only on n_U and n_V . Thus $W \in \mathcal{P}$, proving the claim.

Now let $j : V \hookrightarrow X$ be a quasi-compact open immersion and let $\mathcal{B}$ be a basis of extremely compactly bounded quasi-compact opens. Since V is quasi-compact, there exist finitely many basis elements $B_1, \dots, B_r \in \mathcal{B}$ such that $V = B_1 \cup \dots \cup B_r$. For every non-empty subset $I \subseteq \{1, \dots, r\}$, set $B_I = \bigcap_{i \in I} B_i$ and denote by $j_I : B_I \hookrightarrow X$ the composite open immersion. By above, every B_I belongs to $\mathcal{P}$. Therefore, for each non-empty I , there exists an integer $n_I \in \mathbb{N}$ and a perfect complex $Q_I \in \text{Perf}(X)$ such that $(j_I)_* \mathcal{O}_{B_I} \in \text{Thick}_{n_I}(Q_I[-n_I, n_I])$. Since there are only finitely many non-empty subsets $I \subseteq \{1, \dots, r\}$, there exists a single integer $N \in \mathbb{N}$ and a single perfect complex $Q = \bigoplus_{\emptyset \neq I \subseteq \{1, \dots, r\}} Q_I$ such that $(j_I)_* \mathcal{O}_{B_I} \in \text{Thick}_N(Q[-N, N])$ for every non-empty I .

The finite Čech complex of the cover $V = \bigcup_{i=1}^r B_i$ computes $j_* \mathcal{O}_V$. Explicitly, there is a finite totalization

$$j_* \mathcal{O}_V \simeq \text{Tot} \left[\prod_i (j_i)_* \mathcal{O}_{B_i} \rightarrow \prod_{i < j} (j_{ij})_* \mathcal{O}_{B_i \cap B_j} \rightarrow \dots \rightarrow (j_{1 \dots r})_* \mathcal{O}_{B_1 \cap \dots \cap B_r} \right].$$

All products appearing here are finite, hence finite sums in the stable category. Therefore all the terms of this finite Čech complex belong to $\text{Thick}_N(Q[-N, N])$. Since the totalization is finite, after enlarging N once more it follows that $j_* \mathcal{O}_V \in \text{Thick}_N(Q[-N, N])$ for some integer $N \in \mathbb{N}$. Using the projection formula concludes. $\square$

Our next goal is to show that for a quasi-compact quasi-separated scheme whose derived category of quasi-coherent sheaves is approximable every quasi-compact open immersion is extremely compactly bounded. We begin with a technical observation.

Lemma 3.5. Let X be a quasi-compact quasi-separated scheme and let $j : U \hookrightarrow X$ be the inclusion of a quasi-compact open. Then $\text{hom}_X(j_*(\mathcal{O}_U), -) : \text{QCoh}(X) \rightarrow \text{Sp}$ is right t -exact up to a shift.

Proof. The claim is that there exists an integer $m \in \mathbb{N}$ such that $\mathrm{hom}_X(j_*(\mathcal{O}_U), -)$ sends $\mathrm{QCoh}(X)_{\geq 0}$ into $\mathrm{Sp}_{\geq -m}$. Assume first that X is affine, say $X = \mathrm{Spec}(R)$. Since U is a quasi-compact open, there are $f_1, \dots, f_a \in R$ such that $U = D(f_1) \cup \dots \cup D(f_a)$. For every $1 \leq i \leq a$ let $j_i : D(f_i) \hookrightarrow X$ be the open inclusion. Then $(j_i)_*(\mathcal{O}_{D(f_i)}) \simeq R_{f_i}$ may be realized as a telescope, namely as the cofibre of a map $\bigoplus_{n \in \mathbb{N}} R \rightarrow \bigoplus_{n \in \mathbb{N}} R$. In particular, for every $M \in \mathrm{QCoh}(X)_{\geq 0} \simeq (\mathrm{Mod}_R)_{\geq 0}$ it follows that $\mathrm{hom}_X((j_i)_*(\mathcal{O}_{D(f_i)}), M) \simeq \mathrm{fib}(\prod_{n \in \mathbb{N}} M \rightarrow \prod_{n \in \mathbb{N}} M)$ is in $\mathrm{Sp}_{\geq -1}$. This follows by prestability plus the fact that the t -structure on Sp is compatible with arbitrary products. For every non-empty subset $I \subseteq \{1, \dots, a\}$, set $f_I := \prod_{i \in I} f_i$ and let $j_I : D(f_I) \hookrightarrow X$ be the corresponding principal open immersion. The Čech complex of the cover $U = \bigcup_i D(f_i)$ gives an equivalence

$$j_*(\mathcal{O}_U) \simeq \mathrm{Tot} \left[\prod_{|I|=1} (j_I)_*(\mathcal{O}_{D(f_I)}) \rightarrow \prod_{|I|=2} (j_I)_*(\mathcal{O}_{D(f_I)}) \rightarrow \dots \rightarrow (j_{\{1, \dots, a\}})_*(\mathcal{O}_{D(f_1 \dots f_a)}) \right].$$

Since all products are finite, it follows that $C^p := \bigoplus_{|I|=p+1} (j_I)_*(\mathcal{O}_{D(f_I)}) \simeq \bigoplus_{|I|=p+1} R_{f_I}$. Thus $j_*(\mathcal{O}_U)$ is obtained from the bounded Čech complex $C^0 \rightarrow C^1 \rightarrow \dots \rightarrow C^{a-1}$. For every non-empty I , the object R_{f_I} is a principal localization of R . Hence, by the preceding paragraph, $\mathrm{hom}_X(R_{f_I}, -)$ is right t -exact up to a shift. Therefore $\mathrm{hom}_X(C^p, -) \simeq \prod_{|I|=p+1} \mathrm{hom}_X(R_{f_I}, -)$ is still right t -exact up to a shift. Since the Čech complex is bounded, $j_*(\mathcal{O}_U)$ admits a finite filtration whose graded pieces are shifts of the objects C^p . Applying $\mathrm{hom}_X(-, M)$, for $M \in \mathrm{QCoh}(X)_{\geq 0}$ connective, produces a finite tower whose successive fibres are shifts of the spectra $\mathrm{hom}_X(C^p, M)$. Thus the connectivity can decrease only by a bounded amount depending on a , showing that $\mathrm{hom}_X(j_*(\mathcal{O}_U), -)$ is right t -exact up to a shift in the affine case.

Assume now that X is quasi-compact separated. Let $X = V_1 \cup \dots \cup V_a$ be an affine open cover and for every non-empty subset $I \subseteq \{1, \dots, a\}$ consider the quasi-compact open $V_I = \bigcap_{i \in I} V_i$. Since X is separated, the open $V_I \subseteq X$ is affine (being an intersection of affines). Now apply the preceding case to the quasi-compact open inclusion $i : U \cap V_I \hookrightarrow V_I$ to deduce that $\mathrm{hom}_{V_I}(i_*(\mathcal{O}_{U \cap V_I}), -)$ is right t -exact up to a shift. Notice that here the base change $(j_*(\mathcal{O}_U))|_{V_I} \simeq i_*(\mathcal{O}_{U \cap V_I}) \simeq i_*(\mathcal{O}_{U \cap V_I})$ has been used. Since Zariski descent states that

$$\mathrm{QCoh}(X) \simeq \lim_{I \subseteq \{1, \dots, a\}} \mathrm{QCoh}(V_I), \quad \mathrm{QCoh}(X)_{\geq 0} \simeq \lim_{I \subseteq \{1, \dots, a\}} \mathrm{QCoh}(V_I)_{\geq 0}$$

it follows that

$$\mathrm{hom}_X(j_*(\mathcal{O}_U), -) \simeq \lim_{I \subseteq \{1, \dots, a\}} \mathrm{hom}_{V_I}(i_*(\mathcal{O}_{U \cap V_I}), (-)|_{V_I})$$

is still right t -exact up to a shift (since the limit is finite).

Finally, assume that X is quasi-compact quasi-separated. Let $X = W_1 \cup \dots \cup W_b$ be an affine open cover and for every non-empty subset $I \subseteq \{1, \dots, b\}$ consider the quasi-compact open $W_I = \bigcap_{i \in I} W_i$. Then W_I is open in one of the affines W_i , making W_I separated. In particular, the previous paragraph applies to $i : U \cap W_I \hookrightarrow W_I$, making $\mathrm{hom}_{W_I}(i_*(\mathcal{O}_{U \cap W_I}), -)$ right t -exact up to a shift. Using again Zariski descent shows the claim. $\square$

Extremely compactly bounded morphisms are a necessary condition for approximability.

Proposition 3.6. Let X be a quasi-compact quasi-separated scheme. Assume that $\mathrm{QCoh}(X)$ is approximable. Then every quasi-compact open immersion is extremely compactly bounded.

Proof. Let $j : U \hookrightarrow X$ be a quasi-compact open. Since j_* is right t -exact up to a shift, there exists $k \in \mathbb{N}$ such that $\Sigma^k j_*(\mathcal{O}_U) \in \mathrm{QCoh}(X)_{\geq 0}$ is connective. Since $\mathrm{QCoh}(X)$ is approximable for every integer $m \geq 1$ there exists an exact sequence $P \rightarrow \Sigma^k j_*(\mathcal{O}_U) \rightarrow E$ with $P \in \mathrm{Thick}_n(G[-n, n])$ and $E \in \mathrm{QCoh}(X)_{\geq m}$. Here $G \in \mathrm{Perf}(X)$ is a compact generator and $n \in \mathbb{N}$ is an integer (determined via Remark 2.4). Notice now that it suffices to show that the map $\Sigma^k j_*(\mathcal{O}_U) \rightarrow E$ is null-homotopic, since it will make $j_*(\mathcal{O}_U)$ a retract of $\Sigma^{-k} P$, and hence an object of $\mathrm{Thick}_{kn}(G[-n-k, n+k])$, so that the

projection formula will show the claim (as observed in [Example 3.2](#)). Now [Lemma 3.5](#) produces an integer $w \in \mathbb{N}$ such that $\mathrm{hom}_X(j_*(\mathcal{O}_U), -)$ sends $\mathrm{QCoh}(X)_{\geq m-k}$ into $\mathrm{Sp}_{\geq m-k-w}$. In particular, picking m such that $m-k-w \geq 1$ will prove the claim. $\square$

4. APPROXIMABILITY VIA ZARISKI DESCENT

We now work towards the proof of the main result of these notes. We begin with the technical:

Lemma 4.1. Let X be a quasi-compact quasi-separated scheme. Let $N \in \mathbb{N}$ be an integer and let $G \in \mathrm{Perf}(X)$ be compact. Then there exists $w \in \mathbb{N}$ such that $\mathrm{hom}_X(P, \mathrm{QCoh}(X)_{\geq w}) \in \mathrm{Sp}_{\geq 1}$ for every $P \in \mathrm{Thick}(G[-N, N])$. In particular, w can be chosen ≥ 2 .

Proof. Since X is quasi-compact and quasi-separated, the global sections functor $\Gamma(X, -) : \mathrm{QCoh}(X) \rightarrow \mathrm{Sp}$ is right t -exact up to a shift, in that there exists $d \in \mathbb{N}$ such that $\Gamma(X, \mathrm{QCoh}(X)_{\geq m}) \subseteq \mathrm{Sp}_{\geq m-d}$ for every $m \in \mathbb{Z}$. Since G is perfect, its dual $G^\vee$ is perfect and hence has finite tor-amplitude. In other terms, there exists $a \in \mathbb{N}$ such that $G^\vee \otimes \mathrm{QCoh}(X)_{\geq m} \subseteq \mathrm{QCoh}(X)_{\geq m-a}$ for every $m \in \mathbb{Z}$.

Choose $w \geq d+a+N+1$. Consider first the generators $\Sigma^{-n}G$, with $-N \leq n \leq N$. If $F \in \mathrm{QCoh}(X)_{\geq w}$, then

$$\mathrm{hom}_X(\Sigma^{-n}G, F) \simeq \mathrm{hom}_X(G, \Sigma^n F) \simeq \Gamma(X, G^\vee \otimes \Sigma^n F).$$

Since $\Sigma^n F \in \mathrm{QCoh}(X)_{\geq w+n}$ and $n \geq -N$, it follows that $\Sigma^n F \in \mathrm{QCoh}(X)_{\geq w-N}$. Therefore $G^\vee \otimes \Sigma^n F \in \mathrm{QCoh}(X)_{\geq w-N-a}$ since the t -structure is compatible with tensor products. Applying the cohomological amplitude bound for $\Gamma(X, -)$ gives $\Gamma(X, G^\vee \otimes \Sigma^n F) \in \mathrm{Sp}_{\geq w-N-a-d}$ and by the choice of w , this lies in $\mathrm{Sp}_{\geq 1}$.

Now let

$$\mathcal{S}_w = \{P \in \mathrm{QCoh}(X) \mid \mathrm{hom}_X(P, \mathrm{QCoh}(X)_{\geq w}) \in \mathrm{Sp}_{\geq 1}\}.$$

The previous paragraph shows that $\Sigma^{-n}G \in \mathcal{S}_w$ for every $-N \leq n \leq N$. Notice now that the class $\mathcal{S}_w$ is closed under arbitrary coproducts (which follows since the connective half of the t -structure on spectra is closed under arbitrary products), that it is closed under retracts (which follows since every connective half of a t -structure is closed under retracts) and that it is closed under extensions (which follows since every connective half of a t -structure has this property). It follows that $\mathcal{S}_w$ contains the closure of $\{\Sigma^{-n}G \mid -N \leq n \leq N\}$ under arbitrary coproducts, retracts and extensions, thus proving the claim. If necessary, replacing w by $\max\{w, 2\}$ gives $w \geq 2$. $\square$

We have then the following descent result.

Lemma 4.2. Let X be a quasi-compact quasi-separated scheme. Assume that $X = U \cup V$ is a union of quasi-compact open subschemes with intersection $W = U \cap V$ and denote by

$$\begin{array}{ccc} W & \longrightarrow & U \\ \downarrow & \searrow k & \downarrow i \\ V & \xrightarrow{j} & X \end{array}$$

the various inclusions. Assume that the inclusions are extremely compactly bounded. If $\mathrm{QCoh}(U)$, $\mathrm{QCoh}(V)$ and $\mathrm{QCoh}(W)$ are approximable, then $\mathrm{QCoh}(X)$ is approximable.

Proof. Let $G' \in \mathrm{Perf}(X)$ be a compact generator. Since perfect complexes are bounded below, there exists an integer $A' \in \mathbb{N}$ such that $\Sigma^{A'}G' \in \mathrm{QCoh}(X)_{\geq 0}$ and $\mathrm{hom}_X(G', \mathrm{QCoh}(X)_{\geq A'}) \in \mathrm{Sp}_{\geq 1}$. The claim is then that, after changing the compact generator to G and A if necessary, every connective quasi-coherent sheaf $F \in \mathrm{QCoh}(X)_{\geq 0}$ fits inside an exact sequence $P \rightarrow F \rightarrow E$ with $P \in \mathrm{Thick}_A(G[-A, A])$ and $E \in \mathrm{QCoh}(X)_{\geq 1}$.

Consider now the compact object $G_U = i^*(G') \in \mathrm{QCoh}(U)$. Since G_U is a compact generator (by adjunction and fully-faithfulness of i_*) and since $\mathrm{QCoh}(U)$ is assumed approximable, [\[Nee25, Fact 0.26\]](#) implies the existence of an integer $A_U \in \mathbb{N}$ for which the definition of approximability holds with

the integer A_U . The same observation applied to $G_V = j^*(G')$ and $G_W = k^*(G')$ produces integers $A_V \in \mathbb{N}$ and $A_W \in \mathbb{N}$ for which the same conclusion holds. Let B be the maximum of $\{A_U, A_V, A_W\}$ so that the definition of approximability holds for $\mathrm{QCoh}(U)$, $\mathrm{QCoh}(V)$ and $\mathrm{QCoh}(W)$ with the integer $B \in \mathbb{N}$, as observed in [Remark 2.3](#).

Since the pushforwards i_* , j_* and k_* are extremely compactly bounded, there exist integers $n_U, n_V, n_W \in \mathbb{N}$ such that

$$\begin{aligned} i_*(G_U) &\in \mathrm{Thick}_{n_U}(H_U[-n_U, n_U]), \\ j_*(G_V) &\in \mathrm{Thick}_{n_V}(H_V[-n_V, n_V]), \\ k_*(G_W) &\in \mathrm{Thick}_{n_W}(H_W[-n_W, n_W]), \end{aligned}$$

for some compact objects $H_U, H_V, H_W \in \mathrm{Perf}(X)$. Set now $G := G' \oplus H_U \oplus H_V \oplus H_W \in \mathrm{Perf}(X)$ and notice that it is still a compact generator for $\mathrm{QCoh}(X)$ since it contains a compact generator as a direct summand. If N is the maximum of $\{n_U, n_V, n_W\}$, then the above can be rephrased as

$$\begin{aligned} i_*(G_U) &\in \mathrm{Thick}_N(G[-N, N]), \\ j_*(G_V) &\in \mathrm{Thick}_N(G[-N, N]), \\ k_*(G_W) &\in \mathrm{Thick}_N(G[-N, N]), \end{aligned}$$

as follows from the definition. Finally, let r_i, r_j and $r_k \in \mathbb{N}$ be the errors of right t -exactness of i_* , j_* and k_* . Set r to be the maximum of the three, so that i_* , j_* and k_* will send connective objects to $(-r)$ -connective objects.

Enough with notation, now it is time for the actual proof. Pick a connective object $F \in \mathrm{QCoh}(X)_{\geq 0}$. Since the categories $\mathrm{QCoh}(U)$, $\mathrm{QCoh}(V)$ are approximable by assumption, [Remark 2.4](#) provides exact sequences

$$P_U \rightarrow i^*(F) \rightarrow E_U, \quad P_V \rightarrow j^*(F) \rightarrow E_V$$

in which

$$\begin{aligned} P_U &\in \mathrm{Thick}_{(r+1)B}(G_U[-B-r-1, B+r+1]) \\ P_V &\in \mathrm{Thick}_{(r+1)B}(G_V[-B-r-1, B+r+1]) \end{aligned}$$

where the cofibres $E_U \in \mathrm{QCoh}(U)_{\geq r+1}$ and $E_V \in \mathrm{QCoh}(V)_{\geq r+1}$ are $(r+1)$ -connective. Pushing to X and summing the two exact sequences produces an exact sequence

$$i_*(P_U) \oplus j_*(P_V) \rightarrow i_*i^*(F) \oplus j_*j^*(F) \rightarrow i_*(E_U) \oplus j_*(E_V)$$

in which the first term is a sum of objects in

$i_*(P_U) \in \mathrm{Thick}_{(r+1)B}(i_*(G_U)[-B-r-1, B+r+1]) \subseteq \mathrm{Thick}_{(r+1)BN}(G[-B-r-1-N, B+r+1+N])$
 $j_*(P_V) \in \mathrm{Thick}_{(r+1)B}(j_*(G_V)[-B-r-1, B+r+1]) \subseteq \mathrm{Thick}_{(r+1)BN}(G[-B-r-1-N, B+r+1+N])$
and hence it lives in $\mathrm{Thick}_{(r+1)BN}(G[-B-r-1-N, B+r+1+N])$. On the other side, the cofibre $i_*(E_U) \oplus j_*(E_V)$ lives in $\mathrm{QCoh}(X)_{\geq 1}$ by definition of the error r of right t -exactness. Since the sum $i_*(P_U) \oplus j_*(P_V)$ lives in the above thickening, [Lemma 4.1](#) produces an integer $w \in \mathbb{N}$, assumed $w \geq 2$, such that

$$\mathrm{hom}_X(i_*(P_U) \oplus j_*(P_V), \mathrm{QCoh}(X)_{\geq w}) \in \mathrm{Sp}_{\geq 1}.$$

This $w \in \mathbb{N}$ can now be fed into the definition of approximability of $\mathrm{QCoh}(W)$ to produce an exact sequence

$$P_W \rightarrow k^*(F) \rightarrow E_W$$

in which $P_W \in \mathrm{Thick}_{(w+r)B}(G_W[-B-w-r, B+w+r])$ and, more importantly, for which the cofibre $E_W \in \mathrm{QCoh}(W)_{\geq w+r}$ is $(w+r)$ -connective. Consider now the diagram

$$\begin{array}{ccc} i_*(P_U) \oplus j_*(P_V) & \dashrightarrow & k_*(P_W) \\ \downarrow & & \downarrow \\ F & \longrightarrow & i_*i^*(F) \oplus j_*j^*(F) \longrightarrow k_*k^*(F) \end{array}$$

where the bottom row is the standard Mayer-Vietoris exact sequence. The goal is to produce the dotted map making the diagram commute. This follows since the obstruction to such a lift lives in $\pi_0 \operatorname{hom}_X(i_*(P_U) \oplus j_*(P_V), k_*(E_W))$ which is zero since $k_*(E_W) \in \operatorname{QCoh}(X)_{\geq w}$ by the definition of E_W and the error r of right t -exactness. Thus the above diagram can be completed to a morphism of exact sequences

$$\begin{array}{ccccc} P & \longrightarrow & i_*(P_U) \oplus j_*(P_V) & \longrightarrow & k_*(P_W) \\ \downarrow & & \downarrow & & \downarrow \\ F & \longrightarrow & i_*i^*(F) \oplus j_*j^*(F) & \longrightarrow & k_*k^*(F) \end{array}$$

Let E be the cofibre of $P \rightarrow F$ and notice that the 3×3 -lemma provides the exact sequence $E \rightarrow i_*(E_U) \oplus j_*(E_V) \rightarrow k_*(E_W)$. Since $w \geq 2$, it is the suspension of a 1-connective object, so that $E \in \operatorname{QCoh}(X)_{\geq 1}$ is 1-connective by prestability. It remains to show that P is proxy small. However, rotating the exact sequence defining P shows that it is an extension

$$\Sigma^{-1}k_*(P_W) \rightarrow P \rightarrow i_*(P_U) \oplus j_*(P_V)$$

where the first object lives in

$$\begin{aligned} \Sigma^{-1}k_*(P_W) &\in \operatorname{Thick}_{(w+r)B}(k_*(G_W)[-B-w-r-1, B+w+r+1]) \\ &\subseteq \operatorname{Thick}_{(w+r)BN}(G[-B-w-r-1-N, B+w+r+N+1]) \end{aligned}$$

and the second one in $\operatorname{Thick}_{(r+1)BN}(G[-B-r-1-N, B+r+1+N])$. Let A_0 be the maximum of $\{(w+r)BN, B+w+r+1+N, (r+1)BN, B+r+1+N\}$ so that both terms live in $\operatorname{Thick}_{A_0}(G[-A_0, A_0])$, thus proving that $P \in \operatorname{Thick}_{2A_0}(G[-A_0, A_0])$. To sum up, the construction has produced for every $F \in \operatorname{QCoh}(X)_{\geq 0}$ an exact sequence $P \rightarrow F \rightarrow E$ in which $P \in \operatorname{Thick}_{2A_0}(G[-2A_0, 2A_0])$ and $E \in \operatorname{QCoh}(X)_{\geq 1}$. Finally, if A is the maximum of $2A_0$ and A' , then A will show both conditions in the definition of approximability, thus proving that $\operatorname{QCoh}(X)$ is approximable. $\square$

We deduce the main result of these notes.

Theorem 4.3. Let X be a quasi-compact quasi-separated scheme with a basis of affine extremely compactly bounded opens. Then $\operatorname{QCoh}(X)$ is approximable.

Proof. Choose a finite cover $X = U_1 \cup \dots \cup U_r$ by affine extremely compactly bounded opens. The proof goes by induction on r that $X_m := U_1 \cup \dots \cup U_m$ has approximable category $\operatorname{QCoh}(X_m)$. For $m = 1$ this is the affine case. Assume the claim for X_{m-1} and set $W_m := X_{m-1} \cap U_m$. Since U_m is affine, W_m is quasi-compact separated, hence $\operatorname{QCoh}(W_m)$ is approximable by Neeman's theorem. Moreover, by Lemma 3.4, every quasi-compact open immersion into X is extremely compactly bounded. Hence the inclusions $X_{m-1} \hookrightarrow X$, $U_m \hookrightarrow X$, and $W_m \hookrightarrow X$ are extremely compactly bounded. Applying Lemma 4.2 to the cover $X_m = X_{m-1} \cup U_m$ shows that $\operatorname{QCoh}(X_m)$ is approximable. Taking $m = r$ and using the cancellation property of Lemma 3.3 proves the claim. $\square$

Corollary 4.4. Let X be a quasi-compact quasi-separated scheme. Then $\operatorname{QCoh}(X)$ is approximable if and only if X admits a basis of affine open subschemes whose inclusions into X are extremely compactly bounded.

Proof. The sufficiency is the previous result. Conversely, if $\operatorname{QCoh}(X)$ is approximable, Proposition 3.6 shows that every quasi-compact open immersion into X is extremely compactly bounded. Since X has a basis of affine quasi-compact opens, the claim follows. $\square$

Remark 4.5. Note that the proof of Lemma 4.1 has nothing to do with schemes. Indeed, the proof can be easily adapted to show that a category $\mathcal{P}$ satisfying condition (1) of the definition of weakly approximable category, which is the pullback in $\operatorname{Pr}_{\operatorname{st}}^{L, \omega}$ of a span $\mathcal{C} \rightarrow \mathcal{E} \leftarrow \mathcal{D}$ of approximable categories in which the right adjoints are colimit preserving and extremely compactly bounded (and right t -exact up to a shift) functors, is approximable.

5. EXAMPLES

The obstruction to prove approximability in the non-separated case amounts to produce a basis of extremely compactly bounded open immersions. We first show that these are easy to construct when there are enough global sections.

Definition 5.1. Let X be a quasi-compact quasi-separated scheme. We will say that a quasi-compact open subscheme $U \subseteq X$ is *globally finitely principal* if there exist global sections $s_1, \dots, s_r \in \Gamma(X, \mathcal{O}_X)$ such that $U = D(s_1) \cup \dots \cup D(s_r)$.

Lemma 5.2. Let X be a quasi-compact quasi-separated scheme and let $j : U \hookrightarrow X$ be a globally finitely principal open subscheme. Then j is extremely compactly bounded.

Proof. Write $U = D(s_1) \cup \dots \cup D(s_r)$ with $s_1, \dots, s_r \in \Gamma(X, \mathcal{O}_X)$. For every non-empty subset $I \subseteq \{1, \dots, r\}$, set $s_I := \prod_{i \in I} s_i$ for the product, so that $D(s_I) = \bigcap_{i \in I} D(s_i)$ is the intersection. Let $j_I : D(s_I) \hookrightarrow X$ be the corresponding open immersion.

Since localization along a global section is computed by the telescope, an exact sequence

$$\bigoplus_{n \geq 0} \mathcal{O}_X \xrightarrow{\text{id} - s_I} \bigoplus_{n \geq 0} \mathcal{O}_X \rightarrow (j_I)_*(\mathcal{O}_{D(s_I)}).$$

shows that $(j_I)_*(\mathcal{O}_{D(s_I)}) \in \text{Thick}_2(\mathcal{O}_X[-2, 2])$. Consider now the finite Čech complex

$$j_* \mathcal{O}_U \simeq \text{Tot} \left[\prod_{|I|=1} (j_I)_* \mathcal{O}_{D(s_I)} \rightarrow \prod_{|I|=2} (j_I)_* \mathcal{O}_{D(s_I)} \rightarrow \dots \rightarrow (j_{\{1, \dots, r\}})_* \mathcal{O}_{D(s_1 \dots s_r)} \right]$$

associated to the cover $U = D(s_1) \cup \dots \cup D(s_r)$. Since all products are finite and the totalization is finite, it follows that there exists $N \in \mathbb{N}$ such that $j_*(\mathcal{O}_U) \in \text{Thick}_N(\mathcal{O}_X[-N, N])$, and therefore the projection formula criterion for quasi-compact open immersions of [Example 3.2](#) shows that the open immersion j is extremely compactly bounded. $\square$

The first non-trivial class of examples comes from gluings.

Proposition 5.3. Let Y be a quasi-compact quasi-separated scheme and let $a : U \hookrightarrow Y$ be a quasi-compact open immersion which is extremely compactly bounded. Let $f_i : X_i \rightarrow Y$, for $i = 1, 2$, be morphisms from quasi-compact quasi-separated schemes. Consider the pullback squares

$$\begin{array}{ccc} W_1 & \longrightarrow & X_1 \\ \downarrow & & \downarrow f_1 \\ U & \xrightarrow{a} & Y \end{array} \quad \begin{array}{ccc} W_2 & \longrightarrow & X_2 \\ \downarrow & & \downarrow f_2 \\ U & \xrightarrow{a} & Y \end{array}$$

and assume that there exists an isomorphism $\varphi : W_1 \simeq W_2$ over U . Let

$$\begin{array}{ccc} W & \longrightarrow & X_1 \\ \downarrow & \searrow k & \downarrow j_1 \\ X_2 & \xrightarrow{j_2} & X \end{array}$$

be the resulting gluing. Then the open immersions j_1 and j_2 are extremely compactly bounded.

If, moreover, each X_i admits a basis of quasi-compact opens whose inclusions into X_i are extremely compactly bounded, then X admits a basis of quasi-compact opens whose inclusions into X are extremely compactly bounded. In particular, every quasi-compact open immersion into X is extremely compactly bounded and $\text{QCoh}(X)$ is approximable.

Proof. Since X is obtained by gluing two quasi-compact quasi-separated schemes along a quasi-compact open subscheme, the scheme X is quasi-compact and quasi-separated. The morphisms

$f_i : X_i \rightarrow Y$ agree on the common open W , hence glue to a morphism $f : X \rightarrow Y$. By construction, $W = f^{-1}(U)$, and there exists a pullback square

$$\begin{array}{ccc} W & \longrightarrow & U \\ k \downarrow & & \downarrow a \\ X & \xrightarrow{f} & Y. \end{array}$$

Since a is extremely compactly bounded, there exist an integer $n \in \mathbb{N}$ and a perfect complex $Q \in \text{Perf}(Y)$ such that $a_*(\mathcal{O}_U) \in \text{Thick}_n(Q[-n, n])$. Using the base change for open immersions (that is, $k_*(\mathcal{O}_W) \simeq f^*a_*(\mathcal{O}_U)$) and the fact that pullback preserves compact objects, it follows that $k_*(\mathcal{O}_W) \in \text{Thick}_n(f^*(Q)[-n, n])$. Consider now the standard Mayer-Vietoris exact sequence

$$\mathcal{O}_X \rightarrow (j_1)_*\mathcal{O}_{X_1} \oplus (j_2)_*\mathcal{O}_{X_2} \rightarrow k_*\mathcal{O}_W$$

associated to the open cover $X = X_1 \cup X_2$. Since $\mathcal{O}_X$ is perfect and $k_*\mathcal{O}_W$ belongs to a finite proxy-small closure of a perfect complex, after enlarging the integer we get a perfect complex $H \in \text{Perf}(X)$ (namely $H = \mathcal{O}_X \oplus f^*(Q)$) and an integer $N \in \mathbb{N}$ such that

$$(j_1)_*(\mathcal{O}_{X_1}) \oplus (j_2)_*(\mathcal{O}_{X_2}) \in \text{Thick}_N(H[-N, N]).$$

Since the two summands are retracts of the direct sum, it follows that

$$(j_i)_*(\mathcal{O}_{X_i}) \in \text{Thick}_N(H[-N, N])$$

for $i = 1, 2$. By the projection formula criterion for quasi-compact open immersions, this proves that j_1 and j_2 are extremely compactly bounded.

Assume now that each X_i admits a basis of quasi-compact opens whose inclusions into X_i are extremely compactly bounded. Let $x \in X$ and let $V \subseteq X$ be an open neighbourhood of x . Choose $i \in \{1, 2\}$ such that $x \in X_i$. Then $V \cap X_i$ is an open neighbourhood of x inside X_i , so by assumption there exists a quasi-compact open $B \subseteq V \cap X_i$ containing x such that the inclusion $B \hookrightarrow X_i$ is extremely compactly bounded. Since $j_i : X_i \hookrightarrow X$ is extremely compactly bounded, the composition $B \hookrightarrow X_i \hookrightarrow X$ is extremely compactly bounded by [Lemma 3.3](#). Thus such opens B form a basis of X . Therefore X admits a basis of quasi-compact opens whose inclusions into X are extremely compactly bounded. The final claim follows from [Lemma 3.4](#) and [Theorem 4.3](#). $\square$

Remark 5.4. Let Y be a quasi-compact separated scheme and let $U \subseteq Y$ be a quasi-compact open subscheme. Take $X_1 = X_2 = Y$ and glue the two copies of Y along U to get $X := Y \cup_U Y$. Since Y is separated, the open immersion $U \hookrightarrow Y$ is extremely compactly bounded by [Example 3.2](#). Hence [Proposition 5.3](#) shows that the two open immersions $Y \hookrightarrow X$ are extremely compactly bounded. Moreover, since Y is separated, it has a basis of quasi-compact extremely compactly bounded opens. Therefore X has a basis of quasi-compact extremely compactly bounded opens, and every quasi-compact open immersion into X is extremely compactly bounded. In particular, $\text{QCoh}(X)$ is approximable.

Example 5.5. Let X be the affine line with two origins. Then the previous remark applies and shows that $\text{QCoh}(X)$ is approximable, even though X is not separated. This gives a different proof from the standard one (claimed in [\[JS26, Remark 6.3\]](#) and based on the fact that $\text{QCoh}(X)^\vee$ is a hereditary abelian category).

The above result can be applied to more complicated gluings, for example:

Example 5.6. Let $Y = \mathbb{P}_R^n$ and let $U = D_+(F)$ for a nonzero homogeneous polynomial F . Then the gluing $X = \mathbb{P}_R^n \cup_{D_+(F)} \mathbb{P}_R^n$ is quasi-compact and quasi-separated, and it admits a basis of quasi-compact extremely compactly bounded opens by [Proposition 5.3](#), so that $\text{QCoh}(X)$ is approximable.

The next class of examples comes from schematic quotients.

Proposition 5.7. Let G be a group scheme acting on a quasi-compact quasi-separated scheme Y , and assume that there exists a geometric quotient $q : Y \rightarrow X = Y/G$ in schemes. Assume that X is quasi-compact and quasi-separated. Suppose that there exist finitely many G -stable quasi-compact open subschemes $Y_\alpha \subseteq Y$ for $\alpha \in A$ such that, writing $X_\alpha := q(Y_\alpha)$, the following conditions hold:

- (1) Each X_α is a quasi-compact open subscheme of X , and $X = \bigcup_{\alpha \in A} X_\alpha$.
- (2) Each X_α admits a basis of quasi-compact opens whose inclusions into X_α are extremely compactly bounded (for example, this holds if each X_α is separated).
- (3) For every subset $I \subseteq A$ with $|I| \geq 2$, the intersection $X_I := \bigcap_{\alpha \in I} X_\alpha$ is globally finitely principal as an open subscheme of X .

Then X admits a basis of quasi-compact opens whose inclusions into X are extremely compactly bounded. In particular, every quasi-compact open immersion into X is extremely compactly bounded and $\mathrm{QCoh}(X)$ is approximable.

Proof. Notice that if every $j_\alpha : X_\alpha \hookrightarrow X$ is extremely compactly bounded, then X admits a basis of quasi-compact opens whose inclusions are extremely compactly bounded. Indeed, given $x \in X$ and an open neighbourhood $V \subseteq X$ of x , there exists $\alpha \in A$ such that $x \in X_\alpha$. Then $V \cap X_\alpha$ is an open neighbourhood of x inside X_α . By assumption, there exists a quasi-compact open subscheme $B \subseteq V \cap X_\alpha$ containing x such that the inclusion $B \hookrightarrow X_\alpha$ is extremely compactly bounded. Since $X_\alpha \hookrightarrow X$ is extremely compactly bounded, the composition $B \hookrightarrow X_\alpha \hookrightarrow X$ is extremely compactly bounded by Lemma 3.3. In particular, the final claim follows from Lemma 3.4 and Theorem 4.3.

To show that the inclusions $j_\alpha : X_\alpha \hookrightarrow X$ are extremely compactly bounded, consider the finite Čech complex

$$\mathcal{O}_X \rightarrow \prod_{\alpha \in A} (j_\alpha)_*(\mathcal{O}_{X_\alpha}) \rightarrow \prod_{\{\alpha, \beta\} \subseteq A} (j_{\alpha\beta})_*(\mathcal{O}_{X_\alpha \cap X_\beta}) \rightarrow \cdots$$

associated to the open cover $X = \bigcup_{\alpha \in A} X_\alpha$. Since for $|I| \geq 2$ the intersection $X_I := \bigcap_{\alpha \in I} X_\alpha$ is globally finitely principal, Lemma 5.2 implies that the inclusions $X_I \hookrightarrow X$ are extremely compactly bounded. In particular, there exist, for every such I , an integer $n_I \in \mathbb{N}$ and a perfect complex $Q_I \in \mathrm{Perf}(X)$ such that $(j_I)_*(\mathcal{O}_{X_I}) \in \mathrm{Thick}_{n_I}(Q_I[-n_I, n_I])$. Since there are only finitely many such I , after replacing the integer by a larger one and taking a finite direct sum of the Q_I , there exist a single integer $N \in \mathbb{N}$ and a single perfect complex $Q \in \mathrm{Perf}(X)$ such that $(j_I)_*(\mathcal{O}_{X_I}) \in \mathrm{Thick}_N(Q[-N, N])$ for every I with $|I| \geq 2$.

Now the Čech complex is finite and computes $\mathcal{O}_X$. Since $\mathcal{O}_X$ is perfect, arguing on the finite tower of partial totalizations shows that

$$\prod_{\alpha \in A} (j_\alpha)_*(\mathcal{O}_{X_\alpha}) \in \mathrm{Thick}_N(Q[-N, N])$$

after enlarging N and Q if necessary. Since the product is finite, it is a finite direct sum so that each summand $(j_\alpha)_*(\mathcal{O}_{X_\alpha})$ is a retract of the product, thus proving that $(j_\alpha)_*(\mathcal{O}_{X_\alpha}) \in \mathrm{Thick}_N(Q[-N, N])$. In particular, the projection formula criterion of Example 3.2 shows that every open immersion $j_\alpha : X_\alpha \hookrightarrow X$ is extremely compactly bounded. $\square$

Remark 5.8. Let k be a field and let $p, q \geq 1$. Let $Y := \mathbb{A}_k^{p+q} \setminus \{0\}$ be the $(p+q)$ -dimensional punctured affine space and take coordinates $x_1, \dots, x_p, y_1, \dots, y_q$. For $a, b \geq 1$ let $\mathbb{G}_m$ act by $\lambda \cdot x_i = \lambda^a x_i$ on the first p -coordinates and by $\lambda \cdot y_j = \lambda^{-b} y_j$ on the remaining ones. Let $q : Y \rightarrow X := Y/\mathbb{G}_m$ be the geometric quotient. Consider the open cover of Y given by the two $\mathbb{G}_m$ -stable quasi-compact open subschemes

$$Y_+ := \bigcup_{i=1}^p D(x_i), \quad Y_- := \bigcup_{j=1}^q D(y_j).$$

and let $X_+ := q(Y_+)$ and $X_- := q(Y_-)$ be the image in the quotient. Then X_+ and X_- are quasi-compact open subschemes that cover X , which are furthermore separated. Indeed, X_+ is affine over the quotient of $\mathbb{A}_k^p \setminus \{0\}$ by the weight- a action of $\mathbb{G}_m$, and X_- is affine over the quotient of $\mathbb{A}_k^q \setminus \{0\}$

by the weight- b action of $\mathbb{G}_m$; these quotients are isomorphic to $\mathbb{P}_k^{p-1}$ and $\mathbb{P}_k^{q-1}$, respectively. Notice that $X_+ \cap X_-$ is globally finitely principal inside X . Indeed, if g is the greatest common divisor of a, b , then for every pair (i, j) , the function $t_{ij} := x_i^{b/g} y_j^{a/g}$ is $\mathbb{G}_m$ -invariant, hence descends to a global section $t_{ij} \in \Gamma(X, \mathcal{O}_X)$ and $X_+ \cap X_- = \bigcup_{i,j} D(t_{ij})$. Therefore Proposition 5.7 applies, showing that X admits a basis of quasi-compact opens whose inclusions into X are extremely compactly bounded and, more importantly, that $\mathrm{QCoh}(X)$ is approximable. Notice that this example can be generalized to arbitrary weights.

6. THE STANDARD RECOLLEMENT

We now turn our attention to sheaves with support. We start by recalling some terminology.

Remark 6.1. Recall from [CDH⁺25, Appendix A] that a sequence $\mathcal{C} \rightarrow \mathcal{D} \rightarrow \mathcal{E}$ in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ with vanishing composite is called a *Verdier sequence* if $\mathcal{D} \rightarrow \mathcal{E}$ is a localization with kernel given by $\mathcal{C} \rightarrow \mathcal{D}$. In the presentable case, every Verdier sequence is right split: by definition this means that $\mathcal{D} \rightarrow \mathcal{E}$ admits a right adjoint (which can be shown to be fully-faithful).

The following result is well-known; we include a proof for completeness.

Lemma 6.2. Let X be a quasi-compact quasi-separated scheme and let $j : U \hookrightarrow X$ and $i : Z \hookrightarrow X$ be a decomposition of X into a quasi-compact open U and closed Z . Then there exists a right split Verdier sequence

$$\mathrm{QCoh}_Z(X) \rightarrow \mathrm{QCoh}(X) \xrightarrow{j^*} \mathrm{QCoh}(U)$$

in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$. Furthermore, $\mathrm{QCoh}_Z(X) \in \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}, \omega}$ is compactly generated and the inclusion into $\mathrm{QCoh}(X)$ preserves compact objects.

Proof. Since the right adjoint j_* is fully-faithful, the restriction j^* is a localization. An application of [CDH⁺25, Corollary A.1.10] shows the existence of a Verdier sequence $\ker(j^*) \rightarrow \mathrm{QCoh}(X) \rightarrow \mathrm{QCoh}(U)$ in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ which is automatically right split. Thus the claim will be proved if $\mathrm{QCoh}_Z(X) \simeq \ker(j^*)$. Notice, first of all, that $\pi_n(j^*(F)) \simeq j^* \pi_n(F)$ for every $F \in \mathrm{QCoh}(X)$. This follows since j^* is t -exact. For $(\subseteq)$, let $F \in \mathrm{QCoh}_Z(X)$. Then $\pi_n(F)$ is supported on Z , so that $j^* \pi_n(F) \simeq 0$ in $\mathrm{QCoh}(U)^\vee$ for every $n \in \mathbb{Z}$. But this implies that $\pi_n(j^*(F)) \simeq j^* \pi_n(F) \simeq 0$, and since the t -structure on $\mathrm{QCoh}(U)$ is non-degenerate (see for example [BNP23, Lemma 3.6]) it follows that $j^*(F) \simeq 0$, and hence $F \in \ker(j^*)$. Conversely $(\supseteq)$, if $F \in \ker(j^*)$ then $j^* F \simeq 0$, so that $\pi_n(j^* F) \simeq j^* \pi_n(F) \simeq 0$, implying that $F \in \mathrm{QCoh}_Z(X)$. The last claim follows since $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}, \omega}$ has all limits. $\square$

We deduce the following:

Corollary 6.3. Let X be a quasi-compact quasi-separated scheme and let $j : U \hookrightarrow X$ and $i : Z \hookrightarrow X$ be a decomposition of X into a quasi-compact open U and closed Z . Then there exists a recollement

$$\begin{array}{ccccc} & \xleftarrow{\iota^*} & & \xleftarrow{j^!} & \\ \mathrm{QCoh}_Z(X) & \xleftarrow{\iota_*} & \mathrm{QCoh}(X) & \xleftarrow{j^*} & \mathrm{QCoh}(U) \\ & \xleftarrow{\Gamma_Z} & & \xleftarrow{j_*} & \end{array}$$

Proof. It follows since every right split Verdier sequence extends to a recollement by [CDH⁺25, Proposition A.2.19]. $\square$

We now use the above recollement to deduce topological information on the closed complement.

Proposition 6.4. Let X be a quasi-compact quasi-separated scheme and let $j : U \hookrightarrow X$ and $i : Z \hookrightarrow X$ be a decomposition of X into a quasi-compact open U and closed Z . Then Z is an open subscheme if and only if the pullback i^* induces an equivalence $\mathrm{QCoh}_Z(X) \simeq \mathrm{QCoh}(Z)$.

Proof. Assume first that Z is an open subscheme. Then $X \simeq U \sqcup Z$ as schemes and Zariski descent implies that $\mathrm{QCoh}(X) \simeq \mathrm{QCoh}(U) \oplus \mathrm{QCoh}(Z)$. Under this identification, the kernel of $j^* : \mathrm{QCoh}(X) \rightarrow \mathrm{QCoh}(U)$ identifies with $\mathrm{QCoh}(Z)$, so that $\mathrm{QCoh}_Z(X) = \ker(j^*) \simeq \mathrm{QCoh}(Z)$. Moreover, the restriction of i^* to this kernel is exactly the projection onto the second factor, hence inducing the required equivalence.

Conversely, let $\mathcal{K} := \ker(\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z)$ be the quasi-coherent ideal defining the closed immersion $i : Z \hookrightarrow X$. Since U is quasi-compact, we may choose a finitely generated quasi-coherent ideal $\mathcal{J} \subseteq \mathcal{O}_X$ such that $V(\mathcal{J}) = |Z| = V(\mathcal{K})$. This is possible by [Aut18, Lemma 01PH]. Since $\mathcal{J}$ is finitely generated and $\mathcal{J} \subseteq \sqrt{\mathcal{K}}$, there exists $n \geq 1$ such that $\mathcal{J}^n \subseteq \mathcal{K}$. Assume now that i^* induces an equivalence $i^* : \mathrm{QCoh}_Z(X) \rightarrow \mathrm{QCoh}(Z)$. Since the inverse must be given by i_* , it follows that every object of $\mathrm{QCoh}_Z(X)$ lies in the essential image of i_* .

For every $m \geq 1$, the discrete quasi-coherent sheaf $\mathcal{O}_X/\mathcal{J}^m$ belongs to $\mathrm{QCoh}_Z(X)$, since its restriction to U vanishes, and hence it lies in the essential image of i_* . Since i_* is t -exact, it follows that $\mathcal{O}_X/\mathcal{J}^m$ is the pushforward of a discrete quasi-coherent sheaf on Z . Thus the ideal $\mathcal{K}$ acts trivially on $\mathcal{O}_X/\mathcal{J}^m$, and hence $\mathcal{K} \subseteq \mathcal{J}^m$. Applying this with $m = n$ gives $\mathcal{K} \subseteq \mathcal{J}^n \subseteq \mathcal{K}$ and hence $\mathcal{K} = \mathcal{J}^n$. Applying the same argument with $m = n + 1$ gives $\mathcal{K} \subseteq \mathcal{J}^{n+1} \subseteq \mathcal{J}^n = \mathcal{K}$ and hence $\mathcal{J}^n = \mathcal{J}^{n+1}$. Therefore $\mathcal{K}^2 = \mathcal{J}^{2n} = \mathcal{J}^n = \mathcal{K}$. Thus $\mathcal{K}$ is a finitely generated idempotent quasi-coherent ideal. By [Aut18, Lemma 00EH], $\mathcal{K}$ is locally generated by an idempotent, and therefore the closed subscheme $Z = V(\mathcal{K})$ is open and closed in X . Hence Z is an open subscheme. $\square$

Remark 6.5. Notice that the above claim is false if Z is assumed to be only topologically open. Indeed, let k be a field and let $X = \mathrm{Spec}(k[x]/(x^2))$ and $Z = \mathrm{Spec}(k)$. Then Z is a closed subscheme which is topologically open (since the underlying spaces of X and Z agree), but $\mathrm{QCoh}(Z) \simeq \mathrm{Mod}_k$ is not equivalent to $\mathrm{QCoh}_Z(X) \simeq \mathrm{Mod}_{k[x]/(x^2)}$.

7. APPROXIMABILITY FOR QUASI-COHERENT SHEAVES WITH SUPPORT

We can finally prove the second main result of this note.

Theorem 7.1. Let X be a quasi-compact quasi-separated scheme and let Z be a closed subscheme of X with quasi-compact open complement. Then $\mathrm{QCoh}_Z(X)$ is approximable if and only if the underlying topological space $|Z|$ is open and the open subscheme $W \subseteq X$ with underlying topological space $|Z|$ has approximable category $\mathrm{QCoh}(W)$. In this case, $\mathrm{QCoh}_Z(X) \simeq \mathrm{QCoh}(W)$.

Proof. Assume first that $|Z|$ is open. Let $W \subseteq X$ be the open subscheme with underlying space $|Z|$. Since $|Z|$ is also closed, there exists a decomposition $X \simeq U \sqcup W$. Moreover $\mathrm{QCoh}_Z(X) = \mathrm{QCoh}_W(X)$, since both subcategories are defined by the same support condition. Since $\mathrm{QCoh}(W) \simeq \mathrm{QCoh}_W(X)$ by Proposition 6.4, it follows that $\mathrm{QCoh}_Z(X) \simeq \mathrm{QCoh}(W)$. Since $\mathrm{QCoh}(W)$ is approximable by assumption, one implication follows.

Conversely, assume that $\mathrm{QCoh}_Z(X)$ is approximable. By using again [Aut18, Lemma 01PH], choose a finitely generated quasi-coherent ideal $\mathcal{J} \subseteq \mathcal{O}_X$ such that $V(\mathcal{J}) = |Z|$. Choose a compact generator $G \in \mathrm{QCoh}_Z(X)^\omega$. Since $\mathrm{QCoh}_Z(X)$ is approximable and, by Example 2.6, the relevant t -structure is equivalent to the standard one, [Nee25, Fact 0.26] implies that there exists an integer $A \geq 1$ such that $\mathrm{QCoh}_Z(X)$ satisfies conditions (1) and (3) of Definition 2.2 with respect to the standard t -structure.

Since G is compact, it is perfect as an object of $\mathrm{QCoh}(X)$ by Lemma 6.2. In particular, it has only finitely many nonzero homotopy sheaves $\pi_i(G)$, and these are of finite type. Each $\pi_i(G)$ is supported on $V(\mathcal{J})$, so, since $\mathcal{J}$ is finitely generated, there exists $s \geq 1$ such that $\mathcal{J}^s \pi_i(G) = 0$ for every $i \in \mathbb{Z}$. This implies that every object $E \in \mathrm{Thick}_A(G)$ is such that $\mathcal{J}^{sA} \pi_0(E) = 0$. Indeed, the arbitrary shifts of G clearly satisfy the required vanishing, and arbitrary coproducts of objects with that vanishing again satisfy it (since the t -structure on $\mathrm{QCoh}_Z(X)$ is compatible with coproducts). Furthermore, if an object D is an extension $E \rightarrow D \rightarrow F$ of two objects with the vanishing property, then the long

exact sequence in homotopy implies the claim after summing the indices. Finally, it clearly extends to retracts.

Consider now $F := \mathcal{O}_X/\mathcal{I}^{sA+1}$. Since $j^*\mathcal{I} \simeq \mathcal{O}_U$, it follows that the restriction $j^*F \simeq 0$ vanishes, so that $F \in \mathrm{QCoh}_Z(X)_{\geq 0}$ (the connectivity is by construction). By approximability, there exists an exact sequence $E \rightarrow F \rightarrow D$ with $E \in \mathrm{Thick}_A(G[-A, A])$ and $D \in \mathrm{QCoh}_Z(X)_{\geq 1}$. The long exact sequence then shows that $\pi_0(E) \twoheadrightarrow \pi_0(F) \simeq \mathcal{O}_X/\mathcal{I}^{sA+1}$ is an epimorphism and, since $\mathcal{I}^{sA}\pi_0(E) = 0$, it follows that $\mathcal{I}^{sA}(\mathcal{O}_X/\mathcal{I}^{sA+1}) = 0$. Equivalently, $\mathcal{I}^{sA} \subseteq \mathcal{I}^{sA+1}$, so that $\mathcal{I}^{sA} = \mathcal{I}^{sA+1}$. Therefore $\mathcal{I}^{sA}$ is a finitely generated idempotent quasi-coherent ideal. As before, [Aut18, Lemma 00EH] implies that $V(\mathcal{I}) = V(\mathcal{I}^{sA})$ is open and closed in X , making $|Z|$ open. The first paragraph of the proof shows then that $\mathrm{QCoh}_Z(X) \simeq \mathrm{QCoh}(W)$. $\square$

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